DRESS: A Continuous Framework for Structural Graph Refinement

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Abstract

We introduce DRESS, a deterministic, parameter-free framework that iteratively refines the structural similarity of edges in a graph to produce a *canonical fingerprint*: a real-valued edge vector, obtained by converging a non-linear dynamical system to its unique fixed point. The fingerprint is *isomorphism-invariant* by construction, *numerically stable* (all values lie in $[0, 2]$), *fast* and *embarrassingly parallel* to compute: each iteration costs $\mathcal{O}(m \cdot d_{\max})$ and convergence is guaranteed by Birkhoff contraction. As a direct consequence of these properties, DRESS is provably at least as expressive as the 2-dimensional Weisfeiler–Leman (2-WL) test, at a fraction of the cost ($\mathcal{O}(m \cdot d_{\max})$ vs. $\mathcal{O}(n^3)$ per iteration). We generalize the original equation (Castrillo, León, and Gómez, 2018) to Motif-DRESS (arbitrary structural motifs) and Generalized-DRESS (abstract aggregation template), and introduce Δ-DRESS, which runs DRESS on each vertex-deleted subgraph to boost expressiveness. Δ-DRESS empirically separates all 7,983 graphs in a comprehensive Strongly Regular Graph benchmark, and iterated deletion (Δ^k-DRESS) climbs the CFI staircase, achieving $(k+2)$-WL expressiveness at each depth k.

1 Introduction

A fundamental task in graph analysis is to assign to each graph a compact descriptor that is *canonical* (isomorphic graphs receive the same descriptor), *discriminative* (non-isomorphic graphs receive different descriptors whenever possible), and *cheap to compute*. Existing approaches fall into two broad categories: discrete combinatorial methods, such as the Weisfeiler–Leman (WL) hierarchy [10, 2], which partition vertex tuples into color classes at $\mathcal{O}(n^{k+1})$ cost for level k; and learned representations, such as message-passing neural networks [11, 8], which require labeled training data and offer no worst-case guarantees.

DRESS is a parameter-free, continuous dynamical system on *edges* that converges to a unique fixed point of real-valued structural similarities, producing a canonical fingerprint vector for any graph. The Original-DRESS equation [3] achieves this deterministically at $\mathcal{O}(m \cdot d_{\max})$ per iteration with guaranteed convergence (Birkhoff contraction), numerical stability (all values in $[0, 2]$), and no learnable parameters. As a consequence of its mathematical structure, Original-DRESS is provably at least as expressive as 2-WL (Section 4), at a fraction of the cost ($\mathcal{O}(m \cdot d_{\max})$ vs. $\mathcal{O}(n^3)$).

We present a hierarchy of DRESS variants, building from the concrete to the general:

1. **Original-DRESS** (Section 2): The foundational equation, a parameter-free dynamical system on edges that converges to a canonical fingerprint vector.

2. **Expressiveness beyond 1-WL** (Section 3): Original-DRESS distinguishes graphs that 1-WL cannot, including the prism graph from $K_{3,3}$.

3. **Original-DRESS $\geq$ 2-WL** (Section 4): Original-DRESS is at least as powerful as the 2-dimensional Weisfeiler–Leman test.

4. **Motif-DRESS** (Section 5): A generalization replacing triangle neighborhoods with arbitrary structural motifs; Original-DRESS is the $M = K_3$ special case.
5. **Generalized-DRESS** (Section 6): The most abstract template, opening the aggregation function and norm as additional free parameters.
6. **Δ-DRESS** (Section 7): Runs DRESS on each vertex-deleted subgraph $G \setminus \{v\}$, connecting the framework to the reconstruction conjecture. Δ-DRESS empirically separates all 7,983 Strongly Regular Graphs in a comprehensive benchmark, and iterated deletion (Δ^k-DRESS) climbs the CFI staircase with $(k+2)$-WL expressiveness at each depth k.

Isomorphism Test. Throughout this paper, the **graph fingerprint** is the sorted vector of converged edge values $\mathrm{sort}(d^*)$. Two graphs are declared non-isomorphic if and only if their fingerprints differ element-wise beyond a numerical tolerance ϵ. This applies uniformly to all DRESS variants. An equivalent representation is the **histogram** $h(d^*)$: in the unweighted case all DRESS values lie in $[0, 2]$ (see the convergence proof in Section 2.1) and the convergence tolerance is $\epsilon = 10^{-6}$, so each value maps to one of $\lceil 2/\epsilon \rceil = 2 \times 10^6$ integer bins, yielding a fixed-size bin-count vector that uniquely identifies the same multiset. In general, the number of bins is determined by the convergence tolerance: a tolerance of ϵ over the range $[0, 2]$ gives $\lceil 2/\epsilon \rceil$ bins. For weighted graphs, where values may exceed 2, the bin range is extended accordingly. The histogram is particularly useful when fingerprints from multiple subgraphs must be pooled (see Δ-DRESS, Section 7).

All experiments use a convergence tolerance of $\epsilon = 10^{-6}$; in every case tested, convergence was reached in fewer than 31 iterations.

Related Work. Higher-order GNN architectures that exceed 1-WL expressiveness include k-GNN [8] and PPGN [7]. Subgraph-based approaches such as GNN-AK+ [12] and ESAN [1] achieve higher expressiveness by running message-passing networks on vertex-deleted or vertex-marked subgraphs; Δ-DRESS (Section 7) shares the vertex-deletion strategy but replaces learned message passing with deterministic fixed-point iteration. All of the above are **supervised** methods that require labeled training data and learnable parameters. In contrast, DRESS is a deterministic, unsupervised framework that produces canonical fingerprints via fixed-point iteration, without any learning or parameter tuning.

2 Original-DRESS

The Original-DRESS equation [3] is a parameter-free, non-linear dynamical system that iteratively refines edge values based on common-neighbor aggregation.

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)} \right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{1}$$

where $N[u] = N(u) \cup \{u\}$ denotes the closed neighborhood of u, and $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the vertex norm. The self-similarity $d_{uu} = 2$ is invariant under iteration. This equation converges to a unique fixed point, providing a continuous structural fingerprint for the graph.

Self-loops. Self-loops are added to every vertex before iteration (i.e., the algorithm uses the closed neighborhood $N[u] = N(u) \cup \{u\}$). The self-loop edge (u, u) participates in both the aggregation and the vertex norm; without it, an isolated edge with no common neighbors would produce $\|u\| \cdot \|v\| = 0$, making the iteration undefined.

Initialization. The equation is scale-invariant (degree-0 homogeneous), so the fixed point is independent of the initial scale. Any uniform initialization $d_{uv}^{(0)} = c$ for all edges, with $c > 0$, converges to the same unique fixed point d^*. In practice, $c = 1$ is used.

However, because Original-DRESS aggregates strictly over triangles (common neighbors), it cannot distinguish graphs with identical triangle counts per edge (e.g., Strongly Regular Graphs).

2.1 Convergence

Theorem 1 (Convergence of Original-DRESS)**.** *Let $F\colon \mathbb{R}_+^{|E|} \to \mathbb{R}_+^{|E|}$ be the Original-DRESS update operator. Then F converges to a unique fixed point $d^* \in [0,2]^{|E|}$ for any initial state $\mathbf{d}^{(0)} > 0$.*

Proof. The argument proceeds in three steps:

1. **Scale Invariance (Degree-0 Homogeneity):** The numerator $\sum_{x \in N[u] \cap N[v]} (d_{ux} + d_{xv})$ is positively homogeneous of degree $p = 1$ in $\mathbf{d}$, and each vertex norm $\|u\| = \sqrt{\sum_{x \in N[u]} d_{ux}}$ is homogeneous of degree $q = \frac{1}{2}$. The denominator $\|u\| \cdot \|v\|$ is therefore degree $2q = 1 = p$, making F homogeneous of degree 0: for any $\lambda > 0$, $F(\lambda \mathbf{d}) = F(\mathbf{d})$. The iteration is self-regularizing and cannot diverge or collapse to zero.

2. **Boundedness:** The common neighborhood $N[u] \cap N[v]$ is a subset of both $N[u]$ and $N[v]$, so the numerator satisfies $\sum_{N[u] \cap N[v]} (d_{ux} + d_{xv}) \le \|u\|^2 + \|v\|^2$. Self-loop inclusion ensures $\|u\|, \|v\| > 0$, giving the per-step bound $F(\mathbf{d})_{uv} \le \|u\|/\|v\| + \|v\|/\|u\|$, which equals 2 iff $\|u\| = \|v\|$ (AM–GM). The contraction drives all norms toward equality, so $d^* \in [0,2]^{|E|}$.

3. **Contraction on the Hilbert Projective Metric:** F is a positive, degree-0 homogeneous map on the cone $\mathbb{R}_{>0}^{|E|}$. By Birkhoff's contraction theorem, F is a strict contraction under the Hilbert projective metric $d_H(x,y) = \log\bigl(\max_e x_e/y_e \cdot \max_e y_e/x_e\bigr)$, provided F maps a bounded part of the cone into a strictly smaller part, which follows from the boundedness above. By the Banach fixed-point theorem on $(\mathbb{R}_{>0}^{|E|}/\!\sim,\, d_H)$, the iteration converges to a unique ray, and the boundedness step pins it to a finite vector $d^* \in [0,2]^{|E|}$.

A complete formal verification of the contraction constant is deferred to future work; all empirical tests confirm convergence within 20 iterations. □

3 Original-DRESS Distinguishes Graphs That 1-WL Cannot

We now demonstrate a key expressiveness result: Original-DRESS, the simplest member of the DRESS family, already distinguishes graphs that 1-WL (color refinement) cannot.

Theorem 2 (DRESS distinguishes beyond 1-WL)**.** *There exist graph pairs that 1-WL cannot distinguish but Original-DRESS can. In particular, DRESS distinguishes the prism graph ($C_3 \square K_2$) from the complete bipartite graph $K_{3,3}$, a pair that 1-WL provably cannot separate.*

Proof. Both graphs are 3-regular on 6 vertices with 9 edges. Since all vertices have the same degree, 1-WL assigns a uniform color to every vertex in both graphs after every iteration, and therefore cannot distinguish them [2].

DRESS operates on *edges* and aggregates over common neighbors (triangles). We show that the prism must have at least two distinct edge values at convergence, while $K_{3,3}$ has a single uniform value.

$K_{3,3}$**:** No two adjacent vertices share a common neighbor ($K_{3,3}$ is triangle-free). Hence every edge has the same neighborhood structure, and by symmetry the unique fixed point assigns the same value to all 9 edges. Numerically, $d^*_{K_{3,3}} \approx 1.155$ for all edges.

Prism graph: There are two structurally distinct edge types: *triangle edges* (6 edges forming the two triangular faces; each pair of endpoints shares 1 common neighbor) and *matching edges* (3 edges connecting corresponding vertices of the two triangles; each pair of endpoints shares 0 common neighbors besides self-loops). Suppose for contradiction that all 9 edges converge to the same value d^*. Then:

- Triangle edges: $d^* = \dfrac{4+4d^*}{2+3d^*}$, which gives $d^* = \frac{1+\sqrt{13}}{3} \approx 1.535$.
- Matching edges: $d^* = \dfrac{4+2d^*}{2+3d^*}$, which gives $d^* = \frac{2}{\sqrt{3}} \approx 1.155$.

These are contradictory, so the prism cannot have a uniform fixed point. The converged unique edge values are:

Graph	**Unique values**	**Multiplicities**
Prism	$\{0.922,\ 1.709\}$	3 matching edges, 6 triangle edges
$K_{3,3}$	$\{2/\sqrt{3} \approx 1.155\}$	all 9 edges identical

Since 1-WL cannot distinguish them and DRESS can, DRESS distinguishes beyond 1-WL on this instance. □

Remark. This result holds for Original-DRESS alone, the simplest member of the DRESS family, using only triangle neighborhoods and $\mathcal{O}(m \cdot d_{\max})$ computation per iteration. The generalizations that follow (Motif-DRESS, Δ-DRESS) extend this advantage further to graphs that resist even 2-WL.

4 Original-DRESS Is Provably at Least as Powerful as 2-WL

The previous section showed that DRESS distinguishes *specific* graph pairs that 1-WL cannot. We now prove a stronger, general result: Original-DRESS is at least as powerful as the 2-dimensional Weisfeiler–Leman test.

Lemma 3 (DRESS fixed point respects 2-WL color classes)**.** *Let $G = (V, E)$ be a connected graph and let $\mathcal{C}$ be the stable coloring produced by the 2-dimensional Weisfeiler–Leman test on G. Then the unique DRESS fixed point $\mathbf{d}^*$ is constant on color classes: if $c^*_{uv} = c^*_{u'v'}$ under $\mathcal{C}$, then $d^*_{uv} = d^*_{u'v'}$.*

Proof. Let $\{C_1, \ldots, C_\ell\}$ denote the edge color classes of $\mathcal{C}$, and let $\{P_1, \ldots, P_q\}$ denote the vertex color classes induced by the diagonal ($p^*_u = c^*_{uu}$).

Step 1 (Ansatz). Seek positive values $r_1, \ldots, r_\ell > 0$ such that setting $d_{uv} = r_{c^*_{uv}}$ for every edge (u, v) satisfies the DRESS fixed-point system. Substituting into the fixed-point equation for any representative edge (u, v) with $c^*_{uv} = C_i$:

$$r_i \cdot \rho_{p^*_u} \cdot \rho_{p^*_v} \;=\; \sum_{(C_j, C_k)} n(C_i \mid C_j, C_k)\,(r_j + r_k),$$

where $n(C_i \mid C_j, C_k)$ counts common neighbors $x \in N[u] \cap N[v]$ with $(c^*_{ux}, c^*_{xv}) = (C_j, C_k)$, and $\rho_{p^*_u} = \sqrt{\sum_{C_j} n_u(C_j)\, r_j}$ with $n_u(C_j)$ the number of edges from u in color class C_j.

Step 2 (Well-definedness). The counts $n(C_i \mid C_j, C_k)$ are identical for every edge in C_i, because the stable 2-WL coloring assigns the same color to pairs with identical common-neighbor color-pair multisets. Similarly, $n_u(C_j)$ is the same for all vertices u in the same class P_i, so $\rho_{p^*_u}$ depends only on the vertex class. The reduced system is therefore well-posed: each equation involves only the class values $r_1, \ldots, r_\ell$.

Step 3 (Existence). The reduced system is the DRESS fixed-point system on the *quotient graph* Q whose vertices are the classes $P_1, \ldots, P_q$ and whose edges are the classes $C_1, \ldots, C_\ell$, with multiplicities $n(C_i \mid C_j, C_k)$. The quotient graph is connected (since G is) and positively weighted. By the Birkhoff–Hopf contraction theorem (Theorem 1), the reduced system has a unique positive fixed point $(r_1^*, \ldots, r_\ell^*)$.

Step 4 (Uniqueness lift). The vector $\mathbf{d}$ defined by $d_{uv} = r^*_{c^*_{uv}}$ for all $(u, v) \in E$ is a positive fixed point of the full DRESS system on G (by Steps 1–3). The full system also has a unique positive fixed point by Theorem 1. Therefore $d^*_{uv} = r^*_{c^*_{uv}}$ for all (u, v): the DRESS fixed point is constant on 2-WL color classes. □

Theorem 4 (Original-DRESS is at least as powerful as 2-WL)**.** *Let G and H be two connected graphs. If the stable colorings produced by the 2-dimensional Weisfeiler–Leman test on G and H differ, then the DRESS fingerprints differ:* $\mathrm{sort}(\mathbf{d}^*_G) \neq \mathrm{sort}(\mathbf{d}^*_H)$.

Proof. Let $\mathcal{C}_G$ and $\mathcal{C}_H$ be the stable 2-WL colorings of G and H. By assumption, the multisets of edge color classes differ: either the number of classes differs, or the sizes of matching classes differ.

By Lemma 3, the DRESS fixed point on G satisfies $d^*_{uv} = r^*_{c^*_{uv}}$ for each edge, where $(r_1^*, \ldots, r_\ell^*)$ is the unique positive fixed point of the reduced system on the quotient graph Q_G. Likewise for H with quotient Q_H.

Since $\mathcal{C}_G \neq \mathcal{C}_H$, the quotient graphs Q_G and Q_H are structurally different: they differ in the number of vertex/edge classes, or in the multiplicity counts $n(C_i \mid C_j, C_k)$. Different quotient structures yield different reduced fixed-point systems, which by Birkhoff uniqueness have different solutions. The multisets of fixed-point values

$$\{\{r^*_{c^*_{uv}} : (u, v) \in E(G)\}\} \quad \text{and} \quad \{\{r^*_{c^*_{uv}} : (u, v) \in E(H)\}\}$$

therefore differ, and hence $\mathrm{sort}(\mathbf{d}^*_G) \neq \mathrm{sort}(\mathbf{d}^*_H)$. □

Remark 5. *Section 9.3 shows empirically that Δ^k-DRESS $\geq (k+2)$-WL: each deletion level adds exactly one WL dimension of expressiveness.*

5 Motif-DRESS

Original-DRESS is limited to triangle neighborhoods. What if we use other structural motifs? Motif-DRESS generalizes Original-DRESS by replacing the common-neighbor aggregation with a motif-defined symmetric vertex neighborhood $\mathcal{N}_M(u, v) = \mathcal{N}_M(v, u)$: the set of endpoints of edges in instances of a motif M containing edge (u, v) that are adjacent to u or v, always including u and v themselves, with an optional symmetric weight function $\bar{w} : E \to \mathbb{R}_{>0}$ (with $\bar{w}_e = 1$ for unweighted graphs).

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in \mathcal{N}_M(u,v)} \left(\bar{w}_{ux} \cdot d_{ux}^{(t)} + \bar{w}_{xv} \cdot d_{xv}^{(t)}\right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{2}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} \bar{w}_{ux} \cdot d_{ux}^{(t)}}$, and $\bar{w}_{ab} \cdot d_{ab} = 0$ whenever $(a, b) \notin E$. The symmetric weight function $\bar{w}(e)$ acts as a multiplicative factor, controlling how much structural information flows along each edge. Because the weights appear identically in the numerator and denominator (both are degree-1 in $\bar{w} \cdot d$), uniformly scaling all weights does not change the fixed point; only the relative weights matter.

Different choices of motif M yield different neighborhoods:

- **Triangle ($M = K_3$):** $\mathcal{N}_{K_3}(u, v) = N[u] \cap N[v]$, the closed common neighborhood. This recovers Original-DRESS exactly: $\sum_{x \in N[u] \cap N[v]} \left(\bar{w}_{ux}\, d_{ux} + \bar{w}_{xv}\, d_{xv}\right)$.

- K_4 **clique:** For each pair x, y with $\{u, v, x, y\}$ forming a K_4, $\mathcal{N}_{K_4}(u, v)$ contains u, v, x, and y. Each vertex contributes $\bar{w}_{ux}\, d_{ux} + \bar{w}_{xv}\, d_{xv}$.

- **4-cycle ($M = C_4$):** For each 4-cycle u-x-y-v, $\mathcal{N}_{C_4}(u, v)$ contains u, v, x, and y. Vertex x contributes $\bar{w}_{ux}\, d_{ux}$; vertex y contributes $\bar{w}_{yv}\, d_{yv}$ (cross-terms are zero since $(x, v), (u, y) \notin E$ in the C_4).

5.1 Properties

Complexity: $\mathcal{O}(\text{Motif Extraction}) + \mathcal{O}\big(I \cdot \sum_e |\mathcal{N}_M(e)|\big)$, where I is the number of iterations and $\sum_e |\mathcal{N}_M(e)|$ is the total size of all motif neighborhoods. For Original-DRESS (triangles), this reduces to $\mathcal{O}(I \cdot m \cdot d_{\max})$. For motifs such as 4-cycles or K_4 cliques in sparse graphs, the motif neighborhoods can be significantly smaller, making Motif-DRESS faster than Original-DRESS.

Invariant: $d_{uu} = 2$. The self-similarity $d_{uu} = 2$ is a constant maintained throughout iteration. Since the iteration only updates d_{uv} for $u \neq v$, and the norm $\|u\| = \sqrt{\sum_{x \in N[u]} \bar{w}_{ux} \cdot d_{ux}}$ always includes $d_{uu} = 2$, this is a fixed property of the equation, not a free parameter.

5.2 Expressiveness

All experiments below use the K_4 clique motif. The specific SRG pairs tested below are known to be indistinguishable by 2-WL; each successful distinction therefore demonstrates that Motif-DRESS empirically exceeds 2-WL on these instances.

- **Rook vs. Shrikhande:** Successfully distinguishes this pair of SRGs with parameters $(16, 6, 2, 2)$. The Rook graph ($K_4 \square K_4$) contains K_4 cliques while the Shrikhande graph does not, so the K_4-neighborhood sizes differ per edge.

- **Chang Graphs:** Distinguishes 3 of the 6 pairwise comparisons among the four SRGs with parameters $(28, 12, 6, 4)$: T(8) vs each of Chang-1, Chang-2, and Chang-3. The three Chang graphs are pairwise indistinguishable by Motif-K_4 (all three have identical K_4-neighborhood structure per edge).

5.3 Convergence

The convergence proof for Original-DRESS (Theorem 1) generalizes directly to Motif-DRESS: the three sufficient conditions (degree-0 homogeneity with $p = 2q$, boundedness since non-neighbor terms vanish via $\bar{w}_{ab}\, d_{ab} = 0$ for $(a, b) \notin E$ so the numerator remains bounded by $\|u\|^2 + \|v\|^2$, and Birkhoff contraction on the Hilbert projective metric) hold for any motif neighborhood $\mathcal{N}_M$ and symmetric weight function $\bar{w}$. In the unweighted case $d^* \in [0, 2]^{|E|}$; with non-uniform edge weights, vertex norms may differ and converged values can exceed 2.

6 Generalized-DRESS

Motif-DRESS fixes the aggregation to summation and the norm to the product of geometric means. Generalized-DRESS is the most abstract template, allowing any choice of these components as long as the resulting update rule preserves the convergence guarantees (degree-0 homogeneity, boundedness, and contraction; see Section 2.1). For each edge (u, v):

$$d^{(t+1)} = \frac{f\big(\mathbf{d}^{(t)}, \mathcal{N}, \bar{w}\big)}{g\big(\mathbf{d}^{(t)}, \mathcal{N}, \bar{w}\big)} \tag{3}$$

where $d \equiv d_{uv}$ is the similarity value assigned to edge (u, v), and:

- $\mathcal{N}(u, v)$ is a **symmetric neighborhood operator**, the structural context aggregated for (u, v),
- $\bar{w} : E \to \mathbb{R}_{>0}$ is a **symmetric weight function** ($\bar{w}(u, v) = \bar{w}(v, u)$; $\bar{w} \equiv 1$ for unweighted graphs),
- f is the **aggregation function**,
- g is the **norm function**.

Because $\mathcal{N}$ and $\bar{w}$ are symmetric, f and g receive the same inputs for (u, v) and (v, u), so $d(u, v) = d(v, u)$ holds for every member of the family. For Original-DRESS and Motif-DRESS this follows directly from the equation; in the general case it is guaranteed by the symmetry of the inputs.

Original-DRESS and Motif-DRESS are both special cases: Original-DRESS fixes $\mathcal{N}$ to triangles, $f = \text{sum}$, $g = \|u\| \cdot \|v\|$ (product of geometric means), $\bar{w} \equiv 1$; Motif-DRESS generalizes $\mathcal{N}$ to arbitrary motifs and $\bar{w}$ to non-uniform weights while keeping the same f and g. Generalized-DRESS opens all four parameters, enabling variants such as Cosine-DRESS (cosine similarity aggregation) or Minkowski-r norms.

7 Δ-DRESS: Deletion

Δ-DRESS breaks symmetry by running DRESS on each vertex-deleted subgraph $G \setminus \{v\}$ for every $v \in V$. The Δ-DRESS fingerprint is the multiset of per-vertex DRESS fingerprints, or equivalently a pooled histogram accumulating all converged edge values across all n deletions.

Deleting a vertex from a regular graph produces an irregular subgraph where DRESS can now distinguish structure that was hidden by the uniform regularity.

7.1 Connection to the Reconstruction Conjecture

The multiset $\{\{\text{DRESS}(G \setminus \{v\}) : v \in V\}\}$ is directly analogous to the *deck* in the Kelly–Ulam reconstruction conjecture [6, 9], which posits that graphs with $n \geq 3$ are determined (up to isomorphism) by their multiset of vertex-deleted subgraphs. Δ-DRESS computes a continuous relaxation of this deck.

7.2 Properties

Complexity: $\mathcal{O}(n \cdot I \cdot m \cdot d_{\max})$, where I is the number of iterations per deletion. The computation for each deleted subgraph is entirely independent, making Δ-DRESS embarrassingly parallel.

7.3 Expressiveness

Δ-DRESS empirically distinguishes the following non-isomorphic pairs:

- **Rook vs. Shrikhande:** Successfully distinguished ($\text{SRG}(16, 6, 2, 2)$; confounds 2-WL).
- **Chang Graphs:** Distinguished all 6 pairs among T(8) and the three Chang graphs ($\text{SRG}(28, 12, 6, 4)$; confound 2-WL).
- $2 \times C_4$ **vs.** C_8**:** Successfully distinguished (both 2-regular on 8 vertices).
- **Petersen vs. Pentagonal Prism:** Successfully distinguished (both 3-regular on 10 vertices).

7.4 Iterated Deletion: Δ^k-DRESS

We now define the iterated vertex deletion operator formally.

Definition 6 (Δ^k-DRESS)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$:*

$$\Delta^k\text{-}DRESS(\mathcal{F}, G) = h\left(\bigsqcup_{S \subset V,\ |S|=k} \mathcal{F}(G \setminus S)\right)$$

where $G \setminus S$ is the subgraph induced by $V \setminus S$, $\mathcal{F}(G \setminus S)$ is the converged DRESS edge-value vector, and h maps the pooled edge values into a histogram fingerprint.

At depth $k = 0$, no vertices are deleted and Δ^0-DRESS reduces to Original-DRESS. At $k = 1$, Δ^1-DRESS runs DRESS on each of the n vertex-deleted subgraphs $G \setminus \{v\}$. In general, Δ^k-DRESS enumerates all $\binom{n}{k}$ vertex subsets of size k and collects the resulting fingerprints into a multiset. The histogram h serves a practical purpose: storing the full multiset of per-deletion edge vectors requires $\mathcal{O}\big(\binom{n}{k} \cdot |E|\big)$ space, which quickly becomes prohibitive; the histogram compresses this into a fixed-size bin-count vector of $\lceil 2/\epsilon \rceil$ bins, independent of k.

8 Family Structure

The DRESS variants introduced in this paper form a nested hierarchy with one orthogonal composition operator:

$$\text{Generalized-DRESS} \supset \text{Motif-DRESS} \supset \text{Original-DRESS}$$

$\Delta(\cdot)$ is an **orthogonal wrapper** applicable to any of the above: given any DRESS variant $\mathcal{F}$,

$$\Delta\text{-DRESS}(\mathcal{F}) = \{\{\mathcal{F}(G \setminus \{v\}) : v \in V\}\}$$

This strategy is independent of the choice of $\mathcal{N}$, f, g, and $\bar{w}$. Δ-DRESS breaks symmetry by deletion and generalizes to depth k: Δ^k runs DRESS on all $\binom{n}{k}$ vertex-deleted subgraphs. The experimental relationship between deletion depth and WL expressiveness is detailed in Section 9.3.

9 Experimental Results

All experiments use convergence tolerance $\epsilon = 10^{-6}$. For Δ-DRESS, two fingerprint representations are compared: the **multiset fingerprint** (the full sorted concatenation of all per-deletion DRESS vectors) and the **histogram fingerprint**. The multiset representation preserves all numerical information; the histogram is a fixed-size summary that is faster to compare and more convenient for pooling across deletions.

9.1 Convergence on Real-World Graphs

Table 1 reports convergence on real-world graphs spanning four orders of magnitude in size ($\epsilon = 10^{-6}$, max 100 iterations).

9.2 Strongly Regular Graphs

Strongly Regular Graphs (SRGs) are the canonical hard instances for polynomial-time isomorphism methods: all edges share identical local structure (same degree, same common-neighbor counts), so Original-DRESS (Δ^0) assigns the same value to every edge and produces a uniform

Graph	$\lvert V\rvert$	$\lvert E\rvert$	Iter.	Final δ
Wiki-Vote	8,298	103,689	17	8.31×10^{-7}
Amazon co-purchasing	548,552	925,872	18	6.35×10^{-7}
LiveJournal	4,033,138	27,933,062	30	7.09×10^{-7}
Facebook (KONECT)	59,216,215	92,522,012	26	6.84×10^{-7}

Table 1: DRESS convergence iterations and final residual $\delta = \max_e |d_e^{(t)} - d_e^{(t-1)}|$. Even on graphs with 59 M vertices, convergence requires fewer than 31 iterations. Iteration count grows sub-logarithmically with graph size.

fingerprint. This is expected: SRGs are regular, and the Original-DRESS iteration sees no local variation.

Δ^1-DRESS overcomes this limitation. By running DRESS on each vertex-deleted subgraph $G \setminus \{v\}$, the uniform regularity is broken and structurally distinct edges emerge. We tested Δ^1-DRESS on 7,983 strongly regular graphs from the repository of Krystal Guo [5], spanning three parameter families:

Family	Parameters	Graphs	Separated	Min L^∞
Conference (Mathon)	$(45, 22, 10, 11)$	6	**100%**	4.16×10^{-3}
Steiner S(2,4,28)	$(63, 32, 16, 16)$	4,466	**100%**	1.95×10^{-3}
Quasi-symmetric 2-designs	$(63, 32, 16, 16)$	3,511	**100%**	2.23×10^{-3}

All 7,983 graphs are pairwise distinguished by Δ^1-DRESS. The "Min L^∞" column reports the smallest element-wise maximum difference between the fingerprints of any sampled pair (1,000 random pairs per family). Values around 10^{-3} confirm that separations are genuine, not floating-point artifacts. This was further validated by checking that the unique count remains stable across all rounding precisions from 6 to 14 decimal digits.

Multiset vs. histogram fingerprint. Both representations produce identical separation results on all 7,983 SRGs. The multiset fingerprint preserves full numerical precision and is the canonical representation. The histogram fingerprint, with bin width $\epsilon = 10^{-6}$ over $[0, 2]$, maps each value to one of 2×10^6 integer bins. On these families the two representations agree perfectly.

9.3 CFI Staircase

The Cai–Fürer–Immerman (CFI) construction [2] produces the canonical hard instances for the WL hierarchy: distinguishing $\mathrm{CFI}(K_n)$ from $\mathrm{CFI}'(K_n)$ requires at least $(n-1)$-WL. We tested Δ^k-DRESS for $k = 0, 1, 2, 3$:

Base	$\lvert V\rvert$	WL req.	Δ^0	Δ^1	Δ^2	Δ^3
K_3	6	2-WL	✓	✓	✓	✓
K_4	16	3-WL	×	✓	✓	✓
K_5	40	4-WL	×	×	✓	✓
K_6	96	5-WL	×	×	×	✓
K_7	224	6-WL	×	×	×	×

The pattern is exact: each deletion level adds one WL dimension of expressiveness. Δ^k-DRESS distinguishes $\mathrm{CFI}(K_{k+3})$ (requiring $(k+2)$-WL) and fails on $\mathrm{CFI}(K_{k+4})$ (requiring $(k+3)$-WL), confirming a sharp boundary at $(k+2)$-WL.

9.4 DRESS–WL Dominance Conjecture

The experimental evidence above, together with the proven $\geq$ 2-WL bound (Theorem 4) and the CFI staircase results, motivates the following:

Remark 7 (DRESS–WL Dominance Conjecture)**.** *Δ^k-DRESS $\geq (k+2)$-WL for all $k \geq 0$.*

Several structural properties of DRESS make this conjecture reasonable:

1. **Continuous vs. discrete.** The WL hierarchy produces discrete color partitions, while DRESS produces real-valued vectors. Continuous values carry strictly more information than discrete classes: two edges assigned the same WL color may receive different DRESS values.

2. **Edges vs. vertices.** DRESS operates natively on edges, which is analogous to 2-WL operating on vertex pairs. This gives DRESS a structural advantage over 1-WL (which operates on single vertices) at no additional asymptotic cost.

3. **Non-linear contraction.** The DRESS update is a non-linear ratio (not a linear hash), so it can encode multiplicative relationships between neighborhoods that a linear aggregation cannot.

4. **Deletion $\equiv$ individualization.** Each deletion level of Δ^k-DRESS produces $\binom{n}{k}$ structurally distinct subgraphs, mirroring the individualization mechanism by which $(k+1)$-WL lifts k-WL. The Reconstruction Conjecture asserts that the deck determines the graph, bridging deletion and individualization in discriminative power.

The base case ($k = 0$) is proved (Theorem 4). The inductive step ($k \geq 1$) remains open.

9.5 Complexity

Original-DRESS ($k = 0$) already matches 2-WL, and each deletion level adds one WL dimension of expressiveness. Yet it is cheaper to compute: a single Original-DRESS run costs $\mathcal{O}(I \cdot m \cdot d_{\max})$ where I is the number of iterations, and depth-k requires $\binom{n}{k}$ independent runs, giving a total of $\mathcal{O}\big(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\big)$, compared to $\mathcal{O}(n^{k+3})$ for $(k+2)$-WL. Space complexity is $\mathcal{O}(n+m)$, compared to $\mathcal{O}(n^{k+2})$ for $(k+2)$-WL. The algorithm is embarrassingly parallel in two orthogonal ways: across the $\binom{n}{k}$ subproblems, and across edge updates within each iteration, enabling distributed/cloud plus multi-core/GPU/SIMD implementations.

10 Conclusion

We presented DRESS, a deterministic, parameter-free framework that assigns to any graph a canonical fingerprint vector in continuous space. The fingerprint is isomorphism-invariant, numerically stable ($d^* \in [0,2]^{|E|}$), and computed at $\mathcal{O}(m \cdot d_{\max})$ per iteration with guaranteed convergence (Theorem 1). As a consequence of its mathematical structure, Original-DRESS is provably at least as expressive as 2-WL (Theorem 4), at a fraction of the cost ($\mathcal{O}(m \cdot d_{\max})$ vs. $\mathcal{O}(n^3)$). We generalized the equation to Motif-DRESS (arbitrary structural motifs) and Generalized-DRESS (abstract aggregation template), and introduced Δ-DRESS, which breaks symmetry by vertex deletion. Δ-DRESS empirically separates all 7,983 Strongly Regular Graphs in a comprehensive benchmark, and iterated deletion (Δ^k-DRESS) climbs the CFI staircase with $(k+2)$-WL expressiveness at each depth k. Because DRESS produces a real-valued edge vector with the same structural semantics as WL color refinement, it serves as a drop-in replacement for WL in downstream applications. This has already been demonstrated in community detection [3, 4], where DRESS edge values naturally classify intra- and inter-community edges. An open-source implementation with bindings for C, C++, Python, Rust, Go, Julia, R, MATLAB, and WebAssembly is publicly available at `https://github.com/velicast/dress-graph`.

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